

Use vision-language models for few-shot industrial defect detection to overcome cold-start in new anomaly scenarios.

Manufacturing × Automated quality inspection × Generative AI · OS037 v1 · refreshed 2026-08-18

ATTRACTIVENESS	RIGHT TO WIN	COMPETITION	SERVICEABLE MARKET	HORIZON	PORTFOLIO
24 is the world moving	60 can we play, can we win	HIGH 0 named in the evidence	€611m observed evidence · per year	LATER research and standards activi...	L2 13 signals

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

The opportunity

This opportunity uses vision-language models for few-shot industrial defect detection, enabling manufacturers to overcome the cold-start problem when new anomaly types appear with limited annotated data. It targets quality inspection teams in manufacturing who need to deploy defect detection faster and with less labeled data. The need is urgent because traditional object detectors struggle in few-shot settings, while VLMs like YOLO-World show promise.

Market opportunity

Method	Total (TAM)	Serviceable (SAM)	Obtainable (SOM)	Confidence
Bottom-up: enterprises x adoption x contract value	€852m €153m – €4.6bn	€611m €110m – €3.3bn	€10.7m €1.9m – €57.5m	observed
Observed: tendered contract value, annualised	€20.9m €20.9m – €56.6m	€20.9m €20.9m – €56.6m	€367k €367k – €991k	observed

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

Bottom-up: enterprises x adoption x contract value

Factor	Value	Basis	Source	As of
Enterprises in Manufacturing (EU27_2020)	238,122 enterprises	observed	Eurostat · sbs_sc_ovw	2024
Enterprises adopting Generative AI	6.4 % of enterprises (10+ employees)	observed	Eurostat · isoc_eb_ain2 · E_AI_TNLG	2025
Annual value of one engagement	€337k per enterprise per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-05
Size-weighted buyer base	39,220 enterprise-equivalents at large-enterprise engagement value	assumption	config/sizing.yaml · size_class_value_weights	size-2026-08-a
Assumed obtainable share of the serviceable market	1.8 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Contract values are observed from EU public procurement (TED). For private-sector verticals they are used as a proxy for comparable enterprise deal size: the buying entity differs, the scope of work is comparable. Engagement value is scaled by enterprise size class (10-19: x0.04, 20-49: x0.09, 50-249: x0.35, GE250: x1.0), because the observed contract value comes from large-organisation tenders. 1 of 1 declared geographies are outside the European reference data (TW); the estimate covers the European footprint only.

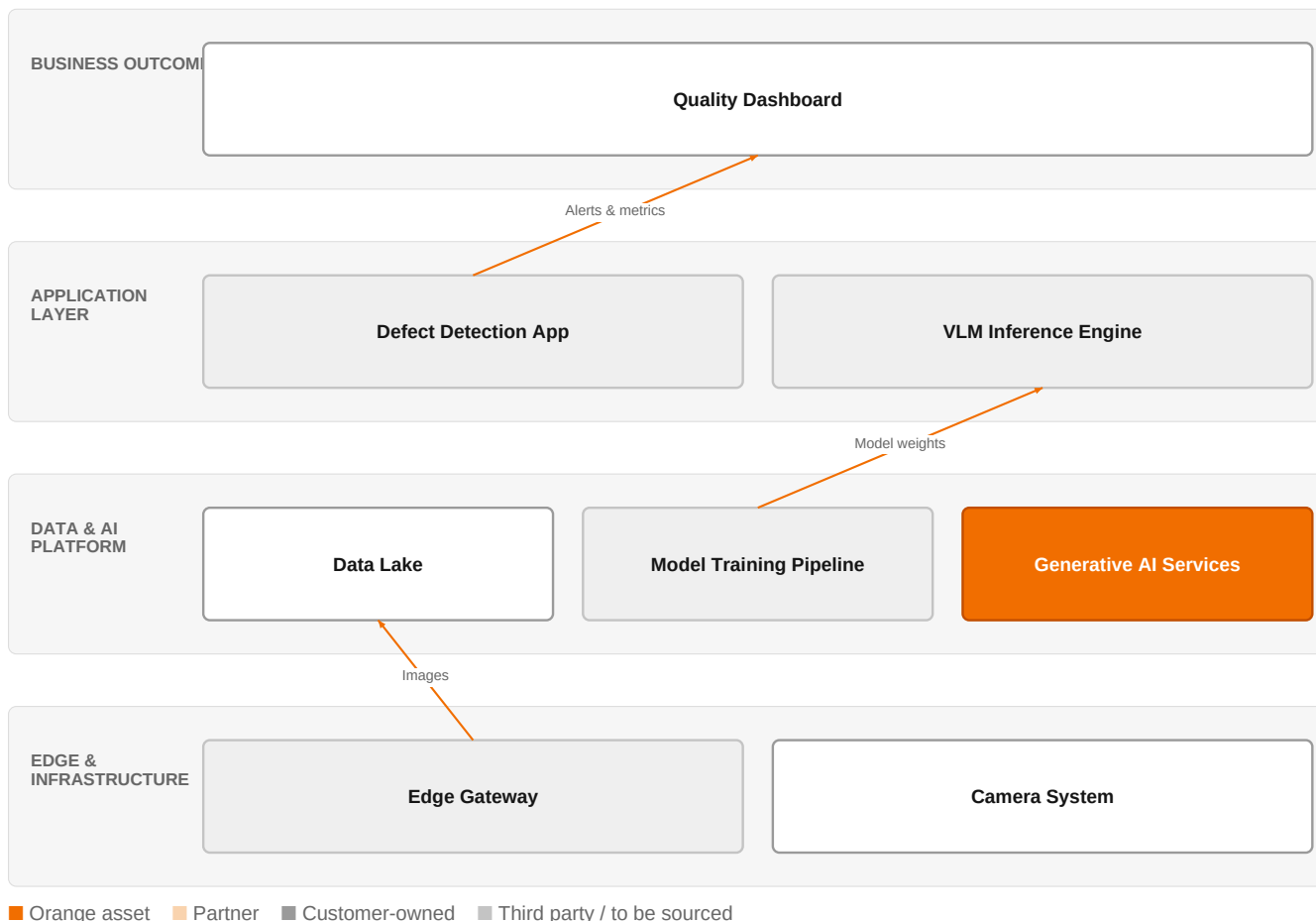
Observed: tendered contract value, annualised

Factor	Value	Basis	Source	As of
Tendered contract value in matching CPV categories	€20.9m per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-05
Assumed obtainable share of the serviceable market	1.8 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Observed over 85 days of TED coverage and scaled to a year (x4.3); a short collection window makes that scaling rough. 14 matching notices, matched on vertical via the CPV crosswalk — §4.5.2's warning applies: a crosswalk error lands directly in this number. Public procurement only. Private-sector demand for the same capability is not visible here at all, so this is a floor rather than a market size. 1 of 1 declared geographies are outside the European reference data (TW); the estimate covers the European footprint only.

The solution, in one picture

Few-Shot Defect Detection Solution



This diagram shows how Orange integrates vision-language models with existing camera infrastructure to enable few-shot defect detection, from edge capture to business dashboard.

Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

What has changed

Recent research demonstrates that adapting vision-language models (VLMs) like YOLO-World can mitigate data scarcity in few-shot industrial defect detection, addressing the cold-start problem where limited annotated data for new anomalies hampers traditional detectors. Additionally, advances in real-time surface defect detection models and the integration of IoT with deep learning for in-situ monitoring are pushing the boundaries of automated quality inspection. These developments signal a shift from manual inspection and traditional deep learning toward more adaptive, data-efficient approaches.

Sources: SIG-8CA4FFD436BC openalex.org 2026-03-27; SIG-92E2E10DA219 openalex.org 2025-12-12; SIG-A06EDB8B4416 arxiv.org 2026-04-22

Who buys, and why now

The buyer is typically a Head of Quality or Manufacturing Operations, while the pain is felt by quality engineers who struggle to label new defect types quickly. The trigger is often a new product line or material where existing defect detection models fail due to lack of training data, causing production delays and increased manual inspection. They need to deploy accurate detection without weeks of data collection and annotation.

Why it is hot now — every claim bound to a dated source

- Adapting vision-language models addresses limited annotated data for new anomalies, a common cold-start problem in automated surface defect detection. [SIG-8CA4FFD436BC]

What Orange would deliver, and why it can win

What Orange would deliver

Orange would assemble a solution leveraging its AI / Data / Cloud experts and IoT experts to integrate vision-language models into the customer's existing camera infrastructure. The engagement would start with a pilot on a specific production line, using Orange's Generative AI capabilities to adapt a VLM to the customer's defect types with minimal labeled examples. Orange would also provide ISO 27001 and TISAX certified infrastructure to ensure data security and compliance, and could involve partners like NVIDIA for GPU acceleration if needed.

Why Orange can win

Orange brings certified infrastructure (ISO 27001, TISAX) that hyperscalers and integrators may not offer, which is critical for manufacturers with strict compliance requirements. Additionally, Orange's IoT expertise and experience with industrial references like Colgate-Palmolive and Saint-Gobain Glass demonstrate domain understanding. However, the core VLM technology is not proprietary to Orange, so the differentiation must come from integration and trust.

Named assets behind this — each individually inspectable (LK-08)

Asset	Type	Link	Why it is linked	Curator
Microsoft	partner	L2	partner provides the required technology	unconfirmed
Google Cloud	partner	L2	partner provides the required technology	unconfirmed
NVIDIA	partner	L2	partner provides the required technology	unconfirmed
Generative AI	technology	L3	no offer path, but adjacent assets or Orange research exist	unconfirmed
ISO 27001	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
TISAX	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
AI / Data / Cloud experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
IoT experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Orange Group researchers	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Colgate-Palmolive	reference	SUP	published reference in this vertical, different use case	unconfirmed
Mondelez International	reference	SUP	published reference in this vertical, different use case	unconfirmed
Saint-Gobain Glass	reference	SUP	published reference in this vertical, different use case	unconfirmed
Skytale	reference	SUP	published reference in this vertical, different use case	unconfirmed

6 further published reference(s) in this vertical are linked to this topic and are listed in the radar.

Competitive landscape

HIGH competitive intensity (score 8.3; 0 of 8 listed players appear in this topic's own evidence)

Crowded, with heavyweight incumbents already visible in the evidence. Win on a specific differentiator or do not bid.

Competitors include hyperscalers like Google Cloud and Microsoft, who offer Generative AI platforms that could be adapted for defect detection, and global system integrators like Accenture, Capgemini, IBM, Infosys, and TCS, who have strong manufacturing practices. These players are strong in AI model development and scaling, but may lack the certified infrastructure and industrial IoT integration that Orange can provide. The customer will likely compare Orange's end-to-end managed service against these alternatives.

Sources: SIG-8CA4FFD436BC openalex.org 2026-03-27

Competitor	Category	Position here	Basis	Evidence
Google Cloud	Hyperscaler	sells Generative AI — also an Orange partner	structural	—
Microsoft	Hyperscaler	sells Generative AI — also an Orange partner	structural	—
Accenture	Global systems integrator	sells Generative AI; active in Manufacturing; competes in OX: Smart Industries	structural	—

Capgemini	Global systems integrator	sells Generative AI; active in Manufacturing; competes in OX: Smart Industries	structural	—
IBM	Global systems integrator	sells Generative AI	structural	—
Infosys	Global systems integrator	sells Generative AI	structural	—
Tata Consultancy Services	Global systems integrator	sells Generative AI	structural	—
Salesforce	Customer-experience platform	sells Generative AI	structural	—

evidenced means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar. **structural** means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register version competitors-2026-08-a.

Qualifying questions for the first meeting

- 1 How many new defect types do you introduce per year, and how long does it currently take to label enough data for a new detection model?
- 2 What is your current false-negative rate on defect detection, and how much does it cost in rework or scrap?
- 3 Do you have a centralized data lake or edge infrastructure where camera images are already collected?
- 4 What compliance certifications (e.g., ISO 27001, TISAX) are mandatory for your quality systems?
- 5 Have you tried using pre-trained vision-language models or generative AI for defect detection? What were the results?

Objections you will meet

They will say	You can answer
We already use a traditional CNN-based detector that works well for our current defects.	That's true for known defects, but the cold-start problem means you need weeks of labeled data for new anomalies. VLM-based few-shot detection can reduce that to days, which is critical when you introduce new products or materials.
We are concerned about data privacy and security if we send images to the cloud.	Orange can deploy the solution on-premises or in a sovereign cloud, and we hold ISO 27001 and TISAX certifications to ensure your data stays protected and compliant.
We already work with a hyperscaler for AI; why should we switch?	We are not asking you to switch. We can complement your existing AI investments by providing the integration, IoT connectivity, and certified infrastructure that hyperscalers often lack. We also have deep manufacturing references like Colgate-Palmolive and Saint-Gobain Glass.

Risks and unknowns

The main risk is that VLM-based few-shot detection may not yet be production-ready for all defect types, as research is still evolving. There is also uncertainty about the performance on customer-specific data, which may require significant customization. Additionally, the competitive landscape is intense, and customers may prefer to work directly with hyperscalers for AI innovation.

Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

Signal	Date	Publisher	T	Title and link
SIG-8CA4FFD436BC	2026-03-27	openalex.org	1	Adapting Vision–Language Models for Few-Shot Industrial Defect Detection
SIG-3B2336BA027B	2026-02-23	openalex.org	1	SynthelTS: Synthetic industrial time-series data with prior knowledge and deep generative model...
SIG-92E2E10DA219	2025-12-12	openalex.org	1	A real-time surface defect detection model based on adaptive feature information selection and ...
SIG-E0AA4F59988C	2025-12-10	openalex.org	1	Anomaly detection for industrial applications: challenges, solutions, and future directions
SIG-A320B94D827D	2025-12-06	openalex.org	1	Anomaly detection model based on unsupervised learning for multivariate industrial time series
SIG-A598AE09F54F	2025-11-20	openalex.org	1	Real-time vision-based defect detection for large-scale on-site earthen additive manufacturing:...
SIG-247B09FE185A	2025-11-07	openalex.org	1	Advancements in Small-Object Detection (2023–2025): Approaches, Datasets, Benchmarks, Applicati...
SIG-E42B61A64931	2025-10-24	openalex.org	1	A time–frequency contrastive learning model for anomaly detection in multivariate time series
SIG-E9AF6D858220	2025-10-21	openalex.org	1	Review on Application of Machine Vision-Based Intelligent Algorithms in Gear Defect Detection
SIG-3C2F7CAC7510	2025-10-01	openalex.org	1	Federated multi scale vision transformer with adaptive client aggregation for industrial defect...
SIG-D46EBCBB2C04	2025-09-16	openalex.org	1	Embedded vision transformers for industrial welding defect detection
SIG-81F8AA75428C	2025-09-09	openalex.org	1	Time-Series-Based Anomaly Detection in Industrial Control Systems Using Generative Adversarial ...
SIG-A06EDB8B4416	2026-04-22	arxiv.org	3	IoT-Enhanced CNN-Based Labelled Crack Detection for Additive Manufacturing Image Annotation in ...

How this brief was made

Computed, not written. Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

Written, then checked. The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

Removed by those checks in this brief: diagram (3 box(es) claimed to be an Orange asset that was not supplied — shown as third-party instead)

What	Version	When
Scores — weight set	w-2026-08-a	2026-08-18T19:08:26+00:00
Market sizing — assumptions	size-2026-08-a	2026-08-18T21:06:15+00:00
Competitor register	competitors-2026-08-a	2026-08-18T21:06:18+00:00
Narrative — prompt / model	describe-v1 / deepseek-chat	2026-08-19T04:56:42+00:00
Pipeline	0.1.0	2026-08-18

Generated 2026-08-19 04:58 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.